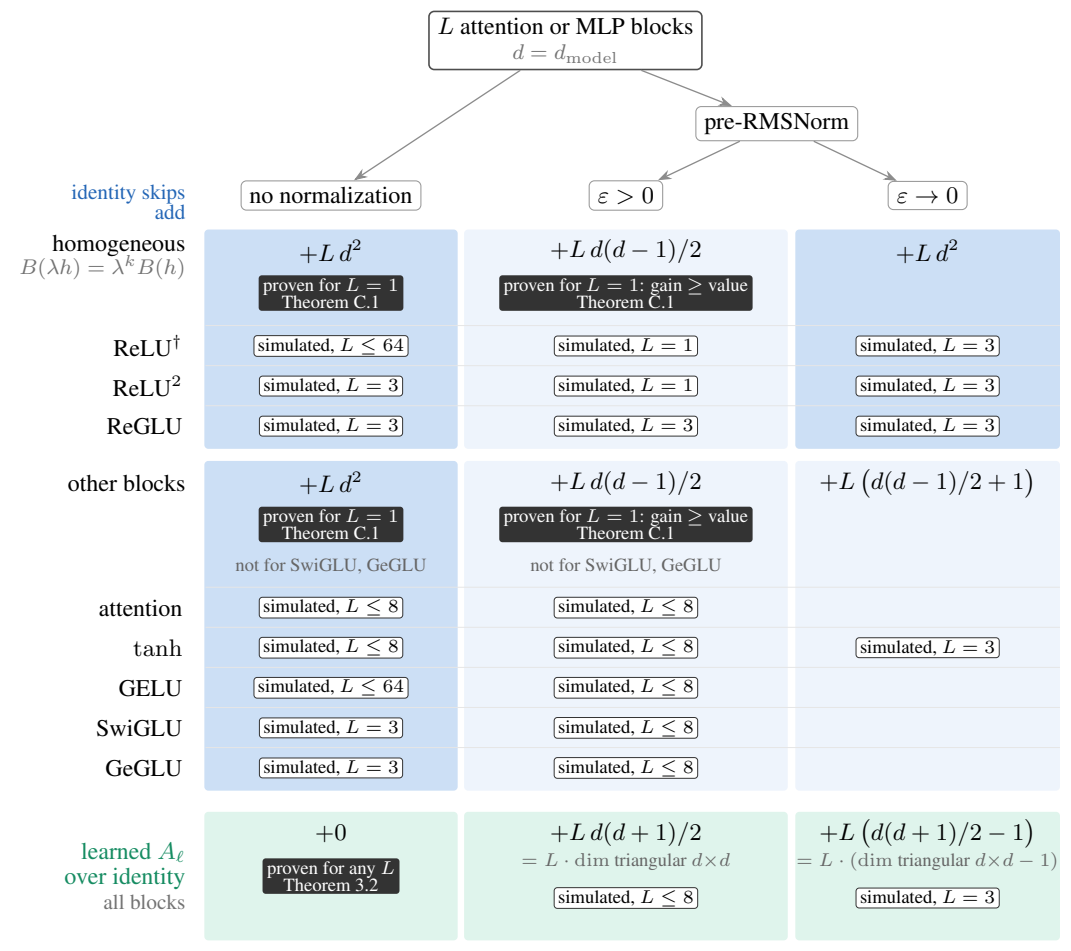

L attention or MLP blocks
$d = d_{\text{model}}$
pre-RMSNorm
identity skips add
no normalization
$\varepsilon > 0$
$\varepsilon \to 0$
homogeneous $B(\lambda h) = \lambda^k B(h)$
$+L\,d^2$
proven for $L = 1$ Theorem C.1
$+L\,d(d-1)/2$
proven for $L = 1$: gain $\geq$ value Theorem C.1
$+L\,d^2$
ReLU†
simulated, $L \leq 64$
simulated, $L = 1$
simulated, $L = 3$
ReLU^2
simulated, $L = 3$
simulated, $L = 1$
simulated, $L = 3$
ReGLU
simulated, $L = 3$
simulated, $L = 3$
simulated, $L = 3$
other blocks
$+L\,d^2$
proven for $L = 1$ Theorem C.1
not for SwiGLU, GeGLU
$+L\,d(d-1)/2$
proven for $L = 1$: gain $\geq$ value Theorem C.1
not for SwiGLU, GeGLU
$+L\left(d(d-1)/2+1\right)$
attention
simulated, $L \leq 8$
simulated, $L \leq 8$
tanh
simulated, $L \leq 8$
simulated, $L \leq 8$
simulated, $L = 3$
GELU
simulated, $L \leq 64$
simulated, $L \leq 8$
SwiGLU
simulated, $L = 3$
simulated, $L \leq 8$
GeGLU
simulated, $L = 3$
simulated, $L \leq 8$
learned A_ℓ over identity
all blocks
$+0$
proven for any L Theorem 3.2
$+L\,d(d+1)/2$
$= L \cdot \dim$ triangular $d \times d$
simulated, $L \leq 8$
$+L\left(d(d+1)/2-1\right)$
$= L \cdot (\dim$ triangular $d \times d-1)$
simulated, $L = 3$